

Master Math for JEE Main & JEE Advanced

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JEE Mains 2020 Jan Chapter wise Question Bank

Heat Transfer

Q1

A Carnot's engine operates between two reservoirs of temperature 900K and 300K. The engine performs 1200 J of work per cycle. The heat energy delivered by the engine to the low temperature reservoir in a cycle is:

7th Jan Morning

Sol

600

$$\eta = \frac{W}{Q_h} = 1 - \frac{300}{900} = \frac{2}{3}$$

$$Q_h = \frac{3}{2} W = 1800 \text{ J}$$

$$Q_L = Q_h - W = 600 \text{ J}$$

Q2

Cascaded Carnot engine is an arrangement in which heat sink of one engine is source for other. If high temperature for one engine is T_1 , low temperature for other engine is T_2 (Assume work done by both engine is same) Calculate lower temperature of first engine.

संयुग्मिक कार्नोट इंजन एक व्यवस्था है जिसमें एक इंजन का ऊष्मीय सिंक दूसरे इंजन के लिए स्रोत की तरह कार्य करता है। यदि एक इंजन का उच्च ताप T_1 है तथा दूसरे इंजन का निम्न ताप T_2 है। (मानिये कि दोनों इंजनों द्वारा किया गया कार्य समान है) तब प्रथम इंजन का निम्न ताप ज्ञात करो।

(1) $\frac{2T_1T_2}{T_1 + T_2}$

(2) $\frac{T_1 + T_2}{2}$

(3) 0

(4) $\sqrt{T_1T_2}$

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(2)

Let, Q_H : Heat input to Ist engine Q_L : Heat rejected from Ist engine Q_L' : Heat rejected from IInd engineWork done by Ist engine = work done by IInd engine

$$Q_H - Q_L = Q_L - Q_L'$$

$$2Q_L = Q_H + Q_L'$$

$$2 = \frac{T_1}{T} + \frac{T_2}{T}$$

$$T = \frac{T_1 + T_2}{2}$$

माना, Q_H : Ist इंजन को दी गई ऊष्मा Q_L : Ist इंजन से निकाली गई ऊष्मा Q_L' : IInd इंजन से निकाली गई ऊष्माIst इंजन द्वारा किया गया कार्य = IInd इंजन द्वारा किया गया कार्य

$$Q_H - Q_L = Q_L - Q_L'$$

$$2Q_L = Q_H + Q_L'$$

$$2 = \frac{T_1}{T} + \frac{T_2}{T}$$

$$T = \frac{T_1 + T_2}{2}$$

Q3

A Carnot engine, having an efficiency of $\eta = 1/10$ as heat engine, is used as a refrigerator. If the work done on the system is 10 J, the amount of energy absorbed from the reservoir at lower temperature is

एक कार्नोट इंजन जिसकी दक्षता $\eta = 1/10$ ऊष्मा इंजन की दक्षता है, रेफ्रिजरेटर के रूप में प्रयोग किया जाता है। यदि निकाय पर किया गया कार्य 10 जूल है, तब न्यून तापमान पर कुण्ड से अवशोषित ऊष्मा की मात्रा है।

(1) 99 J

(2*) 90 J

(3) 1 J

(4) 100 J

8th Jan Evening

Sol

2

For Carnot engine using as refrigerator

कार्नोट इंजन को रेफ्रिजरेटर की भांति प्रयोग करने के लिए

$$W = Q_2 \left(\frac{T_1}{T_2} - 1 \right)$$

$$\text{It is given दिया गया है। } \eta = \frac{1}{10} \Rightarrow \eta = 1 - \frac{T_2}{T_1} \Rightarrow \frac{T_2}{T_1} = \frac{9}{10}$$

So, अतः $Q_2 = 90 \text{ J}$ (as $W = 10 \text{ J}$)

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